9 Cryptography

9.2 Substitution Ciphers

Last time we talked about shift ciphers, and determined that they are not very secure (there are only 26 possible secret keys). What if instead we choose some random mapping, like the following

```
2BQF5WRTD8IJ6HLCOSUVK3A0X9YZN1G4ME7P
ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789
```

Example 9.2.1 Under the above substitution cipher, what is the ciphertext for the message: "March 12 0300"

Example 9.2.2 What are the secret keys for a substitution cipher, and how many secret keys are there?

Unfortunately, this method is still not secure, due to frequency analysis (and even frequency analysis with pairs of letters).

9.3 Transposition Ciphers

Example 9.3.1 Encrypt the message “Meet at first and pine at midnight” using a transposition cipher, and secret key “help”.

Example 9.3.2 Encrypt the message “Lets meet at midnight” using a transposition cipher with secret key “secret”.

Example 9.3.3 The following is an encrypted message using a transposition cipher with secret key "train". What was the message?
`RRSOGRANEI TZAAEOTTVN`

Example 9.3.4 Here is a message which was encrypted using a transposition cipher with secret key "power". What was the message?
`FIYIUHAEFLTNVIUISDCEELRFT`

9.4 Advanced shared symmetric-key methods

Each method so far has been insecure due to frequency analysis. The next method gets around that vulnerability.

Example 9.4.1 Consider an encryption method using a shift cipher, but after each letter is encrypted, the shift is changed by some known amount. The secret key in this system is then the starting shift amount, and how the shift changes between each letter. Let's see an example, where the shift changed by +1 after each letter. Encrypt the phrase "See me".

`2BQF5WRTD8IJ6HLCOSUVK3A0X9YZN1G4ME7P`
`ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789`